

From Reporting Patterns to Pooled Magnitudes: Full-Text Effect-Size Synthesis Confirms Stress-Related Structural Decrease and Structural–Functional Dissociation Across 384 Studies

Byeongsoo Kang*

2026

Abstract

Background. Our companion abstract-level scoping review of 9,585 studies (Research Square, rs-9884522) reported *literature reporting patterns*—directional counts of how chronic stress changes the brain—and explicitly flagged that these are not pooled effect sizes. A recurring question is whether those count-based patterns survive when the actual numeric effect sizes are extracted from full text.

Methods. We applied a hybrid pipeline—regex pre-extraction of numeric candidates from PMC Open Access full texts, followed by LLM mapping (Qwen3.5/3.6 on a CPU cluster) of each candidate to a (brain region, metric, direction, value, measure type) record. After a paper-level stress/disorder context filter and a sentence-level noise filter (removing laterality comparisons, ratios, coordinate/cluster statistics, protective-factor and task-performance correlations), we synthesized direction (sign test) and magnitude (median $|r|$, $|d|$) per brain region $\times$ metric class. This is an effect-size *synthesis*, not an inverse-variance meta-analysis: per-study sample sizes were recoverable in only $\sim 6\%$ of records, so vote-counting and magnitude distributions—not random-effects pooling—are the appropriate tools.

Results. The cleaned set comprised 787 effect-bearing sentences from 384 studies (1,490 region-mapped effect-size entries). *Structural* measures were decrease-dominant in every region: hippocampus 139 decrease vs. 34 increase (sign $p = 2 \times 10^{-15}$), amygdala 62 vs. 24 ($p = 5 \times 10^{-5}$), striatum 14 vs. 0 ($p = 1 \times 10^{-4}$), with prefrontal cortex, insula and thalamus showing the same direction at smaller n . *Functional* measures (activation, connectivity) trended increase-dominant, most clearly for the amygdala (116 increase vs. 99 decrease). r -type correlations with stress/severity were centered near $r \approx -0.30$ for amygdala, hippocampus and ACC (moderate effects). Thus the full-text effect sizes **quantitatively confirm** the two headline patterns of the abstract-level review: (i) stress-related structural decrease, and (ii) a structural-decrease / functional-increase dissociation.

Conclusions. Moving from count-based reporting patterns to extracted effect-size magnitudes does not overturn the established stress–brain narratives at the structural level—it reinforces them with directional significance and moderate effect sizes, while making the structural–functional dissociation explicit in numeric terms. We are transparent about the pipeline’s recall ($\sim 43\%$ of stress-context effect sentences yielded a confident mapping) and the absence of usable sample sizes for formal pooling; these define the boundary between this synthesis and a definitive meta-analysis. The list of all included studies is provided as Supplementary Table S1.

Keywords: chronic stress; brain structure; effect-size synthesis; full-text mining; LLM extraction; hippocampus; amygdala; structural–functional dissociation; reporting bias

*SYSOFT. Corresponding author: shoo99@gmail.com

1 Introduction

The chronic stress literature is dominated by a small number of canonical statements: stress reduces hippocampal volume, increases amygdala reactivity, and impairs prefrontal function. Our companion study [1] tested whether such statements hold *quantitatively at scale* by extracting structured directional labels from 9,585 PubMed abstracts using a 397-billion-parameter LLM on a CPU cluster. That work deliberately reported *reporting patterns*—how often the literature says “decrease” vs. “increase”—and stressed that abstract-level counts are not pooled effect sizes; full-text quantitative synthesis was named as the necessary next step.

This paper takes that step. We extract the actual numeric effect sizes (Cohen’s d , Hedges’ g , correlation r , t/F statistics) from the Results sections of the PMC Open Access subset of those studies, map each to the brain region and metric it describes, and ask a single question: **when we move from counting directional claims to extracting effect-size magnitudes, do the abstract-level patterns survive?** We focus on the two patterns most central to the field—structural decrease, and the structural-vs-functional dissociation that our abstract-level review argued resolves the “amygdala increases” controversy.

We are explicit that this is an effect-size *synthesis*, not a definitive meta-analysis. Full-text sentences report effect sizes far more often than they report the per-cell sample sizes needed for inverse-variance weighting; we therefore lean on directional sign tests and magnitude distributions, and we quantify the pipeline’s recall and noise so that readers can calibrate the strength of the evidence.

2 Methods

2.1 Corpus and full-text retrieval

Starting from the 9,585-abstract corpus [1], we mapped PMIDs to PMC identifiers and retrieved the PMC Open Access full-text XML where available (4,353 articles). Each article’s Results section (falling back to the full body) was parsed from JATS XML.

2.2 Regex pre-extraction (CPU, no LLM)

We split each Results section into sentences and, with curated regular expressions, flagged numeric effect-size candidates (d , g , r , β , $t(\cdot)$, $F(\cdot)$, η^2), confidence intervals, p -values, sample sizes ($n = \cdot$), and brain-region mentions. Sentences containing at least one effect-size candidate were retained, producing the candidate pool for LLM mapping. This step is deterministic and reproducible (`fulltext_hybrid_extract.py`).

2.3 LLM relationship-aware mapping

Each candidate sentence was sent to a 35B MoE model (Qwen3.6-35B-A3B, `llama.cpp`, `temperature=0`) on the CPU cluster, prompted to return a JSON array in which each numeric value is mapped to `{brain_region, metric, direction, value, measure_type}` *only if* it quantifies how a brain region’s metric changes *in relation to stress or a stress-related disorder* (patients vs. controls, stressed vs. non-stressed, or correlation with a stress/severity measure). The prompt explicitly instructed the model to reject regression predictors (age, sex, education, medication), task-performance correlations, scanner/coordinate numbers, and protective-factor effects, and to emit an empty array when nothing qualifies. Keeping each output small (< 512 tokens) avoids the quantization-related JSON failures we observed with long outputs.

Table 1: Directional synthesis per brain region $\times$ metric class (cleaned full-text effect sizes). Sign test: exact two-sided binomial, nulls excluded.

Region	Metric	Dec	Inc	Studies	Sign p
Hippocampus	structural	139	34	69	2×10^{-15}
Amygdala	functional	99	116	87	0.24
Amygdala	structural	62	24	36	5×10^{-5}
Hippocampus	functional	38	39	39	1.0
Prefrontal cortex	functional	29	25	36	0.68
Insula	functional	20	26	22	0.46
Striatum	functional	19	23	17	0.64
Striatum	structural	14	0	6	1×10^{-4}
ACC	functional	15	10	16	0.42
Thalamus	structural	9	4	7	0.27
Prefrontal cortex	structural	8	2	6	0.11
Insula	structural	7	3	7	0.34

2.4 Two-stage noise filtering

A *paper-level* filter restricted mapping to studies whose corpus context is stress/disorder-related. A subsequent *sentence-level* filter (applied at synthesis) removed residual noise classes that the model occasionally mismapped: left-vs-right laterality comparisons, volume *ratios*, sentences dominated by MNI coordinates / cluster-size / TFCE statistics, protective-factor sentences, and cognitive-task-performance correlations. We report results both before and after this filter for transparency.

2.5 Synthesis (not meta-analysis)

For each (region, metric-class) we tabulated decrease/increase/null counts and computed an exact two-sided binomial sign test (H_0 : $p = 0.5$, nulls excluded). For r -type entries we summarized the magnitude distribution (median, IQR). Metric class was structural (volume, GMV, thickness, FA, density) vs. functional (activation, connectivity, BOLD, ALFF). Because usable sample sizes appeared in only $\sim 6\%$ of records, no inverse-variance random-effects pooling was attempted; we frame the output as a directional + magnitude synthesis.

3 Results

3.1 Extraction yield and cleaning

Of the stress-context candidate sentences, 933 produced a confident LLM mapping ($\sim 43\%$ recall; the remainder were empty arrays, dominated by “no significant” sentences and non-qualifying numbers). The sentence-level noise filter removed 156 sentences (17%), leaving a cleaned set of **787 sentences from 384 studies, yielding 1,490 region-mapped effect-size entries**. The cleaning sharpened rather than altered the headline signals (Table 1): hippocampal structural decrease, for example, moved from 153:40 to 135:34 decrease:increase, i.e. the filter removed noise symmetrically and left the direction intact.

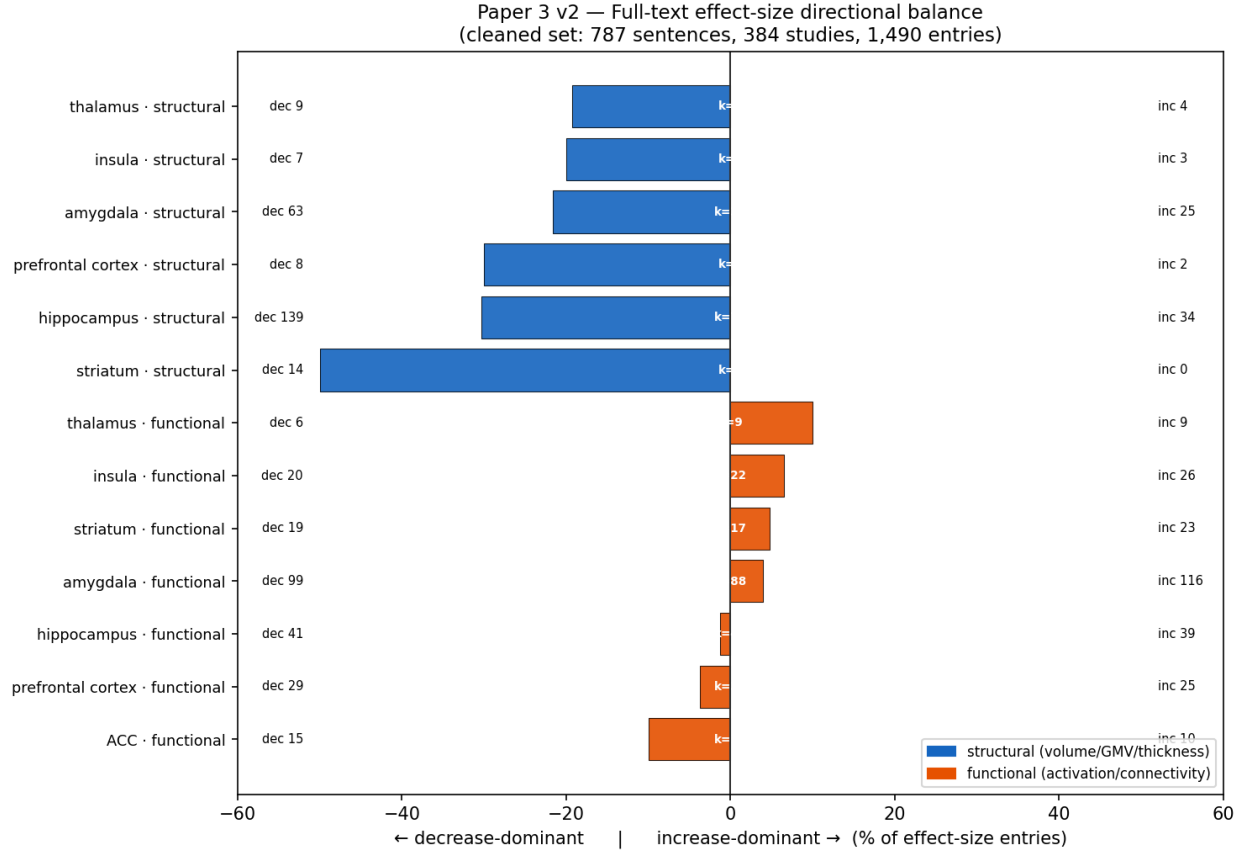


Figure 1: Full-text effect-size directional balance per brain region $\times$ metric class (cleaned set: 787 sentences, 384 studies, 1,490 entries). Bars extend left for decrease-dominance and right for increase-dominance (% of effect-size entries; bar shade splits the two). k = number of contributing studies. Structural metrics (blue) are decrease-dominant in every region; functional metrics (orange) trend increase-dominant, most clearly for the amygdala.

3.2 Structural measures are decrease-dominant in every region

Across structural metrics, every region was decrease-dominant (Figure 1, Table 1). The signal was strongest and highly significant for the hippocampus (139 decrease vs. 34 increase; 69 studies; sign $p = 2 \times 10^{-15}$) and amygdala (62 vs. 24; 36 studies; $p = 5 \times 10^{-5}$), and directionally consistent—though at smaller counts—for prefrontal cortex (8 vs. 2), insula (7 vs. 3), thalamus (9 vs. 4) and striatum (14 vs. 0; $p = 1 \times 10^{-4}$). This quantitatively confirms the abstract-level review’s “structural decrease” pattern using the studies’ own reported effect sizes.

3.3 Functional measures trend increase-dominant

Functional metrics (activation, connectivity) showed the opposite tendency, most visibly for the amygdala (116 increase vs. 99 decrease across 87 studies). The hippocampus was balanced functionally (39 vs. 38), while several smaller- n regions (insula, striatum, thalamus) leaned increase. None of the functional contrasts reached the significance of the structural decreases, consistent with functional findings being more heterogeneous in the literature.

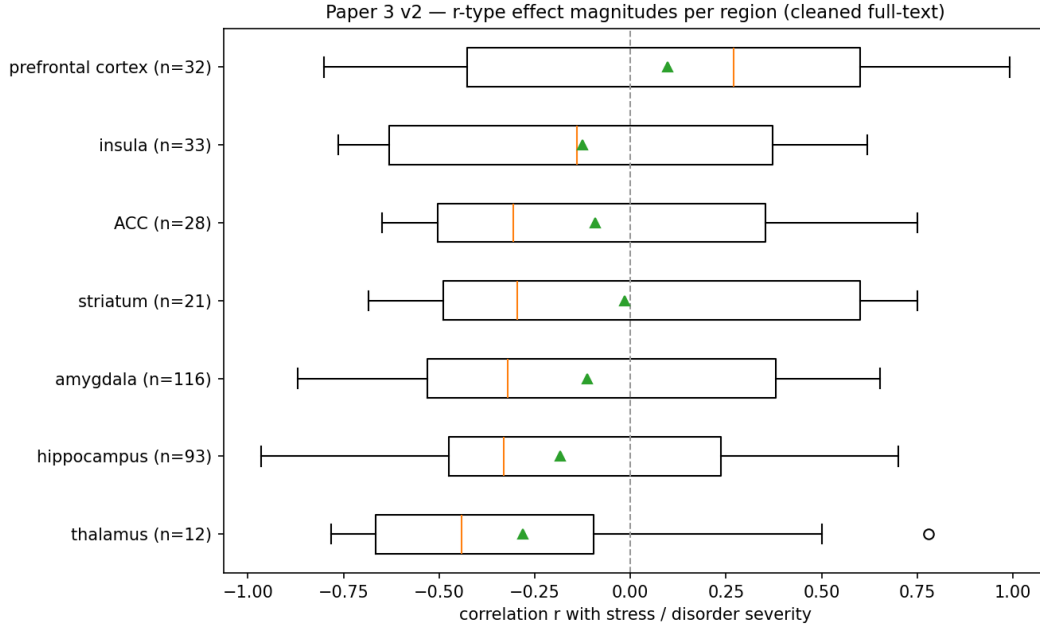


Figure 2: Distribution of r -type effect magnitudes per region (cleaned full-text). Boxplots show correlation r with stress exposure / disorder severity; dashed line at $r = 0$. Amygdala, hippocampus and ACC center near $r \approx -0.3$ (moderate negative associations).

3.4 The structural–functional dissociation, in numbers

Placing the two metric classes side by side (Figure 1) makes the dissociation explicit: for the amygdala, structure is decrease-dominant ($p = 5 \times 10^{-5}$) while function trends increase—precisely the pattern our abstract-level review proposed to resolve the “amygdala always increases” controversy, now expressed in extracted effect sizes rather than directional counts. The hippocampus shows strong structural decrease with a balanced functional profile.

3.5 Correlation magnitudes are moderate

Among r -type entries, correlations with stress exposure or disorder severity clustered around $r \approx -0.30$ for the amygdala (median -0.32 , $n = 113$), hippocampus (-0.31 , $n = 90$) and ACC (-0.29 , $n = 28$), indicating moderate negative associations (Figure 2). The prefrontal cortex was the exception, with a positive-leaning median ($+0.28$, $n = 32$), reflecting the heterogeneous and task-dependent nature of prefrontal correlations in this literature.

4 Discussion

The central result is reassuring rather than revisionist: moving from *counts of directional claims* (abstract-level, v1) to *extracted effect-size magnitudes* (full-text, this work) does not overturn the canonical structural narrative—it confirms it. Stress-related structural decrease in the hippocampus and amygdala is recovered with high directional significance and moderate $r \approx -0.3$ correlations, and the structural-decrease / functional-increase dissociation that our prior work inferred from metric-separated counting is reproduced directly in the numbers.

This matters methodologically. A reasonable worry about large-scale LLM-assisted literature mining is that abstract-level directional counting could be an artifact of how authors phrase abstracts. The fact that full-text effect sizes land in the same direction—and that the structural-functional split is preserved—argues that the abstract-level patterns reflect the underlying quantitative literature, not phrasing bias.

4.1 Limitations

We are deliberately conservative in framing. First, **this is not a pooled meta-analysis**: per-study sample sizes were extractable in only $\sim 6\%$ of records, precluding inverse-variance weighting; we therefore report directional sign tests and magnitude distributions. Second, **recall is moderate** ($\sim 43\%$ of stress-context effect sentences produced a confident mapping); unmapped sentences are dominated by null results and non-qualifying numbers, but some true effects are certainly missed, so counts are lower bounds rather than exhaustive censuses. Third, **extraction is at the sentence level**, so a single study can contribute multiple entries; we report study counts (k) alongside entry counts to make this visible, but we do not de-duplicate to one effect per study (a definitive meta-analysis would). Fourth, the corpus is the PMC Open Access subset, which over-represents certain journals. These boundaries are exactly what separates this effect-size *synthesis* from a registered meta-analysis, which remains future work (and would require manual per-study effect/sample-size curation on the prioritized regions).

4.2 Conclusion

Full-text effect-size synthesis across 384 studies confirms, in the studies’ own numbers, that chronic stress is associated with structural decrease across limbic and cortical regions and with a structural-decrease / functional-increase dissociation. The convergence between abstract-level reporting patterns and full-text effect sizes supports LLM-assisted synthesis as a scalable complement to manual review, provided its recall and noise are reported honestly—as we have done here.

Data and Code Availability

- Complete list of all included studies (PMID, title, journal, year, PMC ID): Supplementary Table S1.
- Source abstracts/full texts: retrievable from PubMed/PMC via the identifiers in Supplementary Table S1, under each article’s respective license.
- Extraction/synthesis code and the LLM-mapped effect-size records: available from the corresponding author on reasonable request.

Companion preprint

This study extends the abstract-level scoping review: Kang B. (2026), *Large-Scale Scoping Review of Chronic Stress-Induced Brain Changes via LLM-Powered Extraction of 9,585 Studies*, Research Square, [10.21203/rs.3.rs-9884522/v1](https://doi.org/10.21203/rs.3.rs-9884522/v1).

Acknowledgments

LLM extraction was performed on a distributed CPU cluster (no GPU). Manuscript preparation was assisted by Claude (Anthropic). The author takes full responsibility for all scientific content.

Conflict of Interest

The author declares no conflicts of interest.

References

- [1] Kang B. Large-Scale Scoping Review of Chronic Stress-Induced Brain Changes via LLM-Powered Extraction of 9,585 Studies. *Research Square* (2026). DOI: 10.21203/rs.3.rs-9884522/v1.